

1. A workers' union is on strike for higher wages. The union claims that the mean salary for teachers is at most Birr 8,400 per year. The legislator does not want to reject the union's claim, however, unless the evidence is very strong against it. Assume that salaries follow a normal distribution and the population standard deviation is known to be Birr 3000. A random sample of 64 teachers is obtained, and the sample mean is Birr, 9,400.

Required: Test if the state legislator accepts the unions' claim at 1% significance level.

2. A fertilizer company claims that the use of its product will result in a yield of at least 35 quintals of wheat per hectare, on average, Application at the fertilizer to a randomly selected 36 sample hectares resulted in a yield of 34 quintals per hectare. Assume the population standard deviation is 5 quintals and yields per hectare are normally distributed.

Required: Test the company's claim at 10% level of significance.

3. A bridge stone claim that its tires have a mean life is 35,000 miles. A random sample of 81 of these tires is tested if the sample mean is 33,000 miles. Assume that the population standard deviation is 3000 miles and the lives of tires are approximately normally distributed.

Required: Test the company's claim using a 5% level of significance.

4. Assume MOHA manufactures and distributes three types of Soft drink: Miranda and Pepsi. In an analysis of the market segments for the two Soft drinks, the firm's market research group raised the question of whether preferences for the two Soft drinks differ among male and female drinkers.

GENDER	SOFT DRINK PREFERENCE	
	Mirinda	Pepsi
Male	20	20
Female	30	10

Required: perform chi –square independent test at 5% level of signfinace (hint $X_{0.05}=4.66$)

5. Assume that a shoe manufacturer company sold 20 pairs of shoes. There are four sizes of shoes and the company expects the size of shoes are sold in equal proportion that is each size is sold 5 times. In the survey conducted the actual number of frequency each size

sold in sequence is as follows; 4, 7, 2, 8. A comparison of observed frequency and expected frequency is tabulated below;

Observed frequency	4	7	2	8
Expected frequency	5	5	5	5

Required: Do goodness –fit test at 5 % level of significance (hint $X_{0.05}=8.12$)

6. **Illustration:** A researcher wishes to try three different techniques to increase the marks of students. The subjects are randomly assigned to three groups; the first group takes accounting, the second group Management, and the third group Business students. After four weeks, the raise in each person's marks is recorded. The data are shown below.

Accounting	management	Business
10	16	15
12	18	19
19	13	12
15	12	18
$\bar{X}_1 = 14$	$\bar{X}_2 = 15$	$\bar{X}_3 = 16$
$S_1^2 = 5$	$S_2^2 = 10$	$S_3^2 = 11$

Required: do F-test value based on ANOVA at $T_{\text{tabulated}} = 4.556$